

COMMERCIAL DATABASES: ORACLE, POSTGRES, MYSQL, NOSQL

1. Oracle Database

- A commercial Relational Database Management System (RDBMS) developed by Oracle Corporation
- Stores data in tables (rows and columns)
- Uses SQL

Key Features

- Strong ACID properties (reliable transactions)
- High security
- Excellent concurrency control
- Advanced features:
 - Partitioning
 - Real Application Clusters (RAC)
 - Backup & recovery tools

Advantages

- Best for **large-scale enterprise systems**
- Highly reliable and secure
- Handles heavy workloads efficiently

Disadvantages

- **Very expensive (license cost)**
- Complex to install and manage
- Requires experienced database administrators

Use Cases

- Banking systems

- Enterprise Resource Planning (ERP)
- Government systems

2. PostgreSQL

- A powerful **open-source RDBMS**
- Known for advanced features and flexibility

Key Features

- Strong **SQL compliance**
- Supports:
 - Complex queries
 - JSON (semi-structured data)
 - Custom functions and extensions
- High **data integrity**

Advantages

- Free and open source
- Highly reliable and stable
- Suitable for complex applications

Disadvantages

- Can be complex for beginners
- Slightly slower for simple read-heavy applications compared to MySQL

Use Cases

- Data analytics
- Enterprise applications
- Geographic Information Systems (GIS)

3. MySQL

- A widely used **open-source RDBMS**
- Very popular in web development

Key Features

- Simple and easy to use
- Fast performance for read operations
- Works well with web technologies

Advantages

- Easy to learn and implement
- Fast for web-based applications
- Large community support

Disadvantages

- Limited advanced features
- Not ideal for very complex queries
- Some advanced features are only available in paid versions

Use Cases

- Websites and blogs
- E-commerce platforms
- Content management systems

4. NoSQL

- A **non-relational database system**
- Does not rely on tables like RDBMS

Types

- Document-based (e.g., MongoDB)
- Key-value (e.g., Redis)

- Column-family (e.g., Cassandra)
- Graph databases (e.g., Neo4j)

Key Features

- Schema-less (flexible structure)
- Designed for large-scale and unstructured data

Advantages

- Highly scalable (horizontal scaling)
- Flexible data storage
- Suitable for big data and real-time applications

Disadvantages

- No standard query language
- Weaker ACID guarantees (in many systems)
- Possible data consistency issues

Use Cases

- Social media platforms
- Real-time analytics
- Big data applications

Comparison Table

Feature	Oracle	PostgreSQL	MySQL	NoSQL
Type	RDBMS	Open-source RDBMS	RDBMS	Non-relational
Cost	Expensive	Free	Free (basic)	Mostly free
Complexity	High	Medium–High	Low	Medium
Performance	Best for heavy workloads	Best for complex queries	Best for simple web apps	Best for large distributed data
Scalability	Vertical	Vertical	Vertical	Horizontal
Schema	Fixed	Fixed	Fixed	Flexible
ACID Support	Strong	Strong	Strong	Weak (many systems)
Use Case	Enterprise systems	Analytics, enterprise	Websites	Big data, real-time